



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to Sakthi Fuels Unit II

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

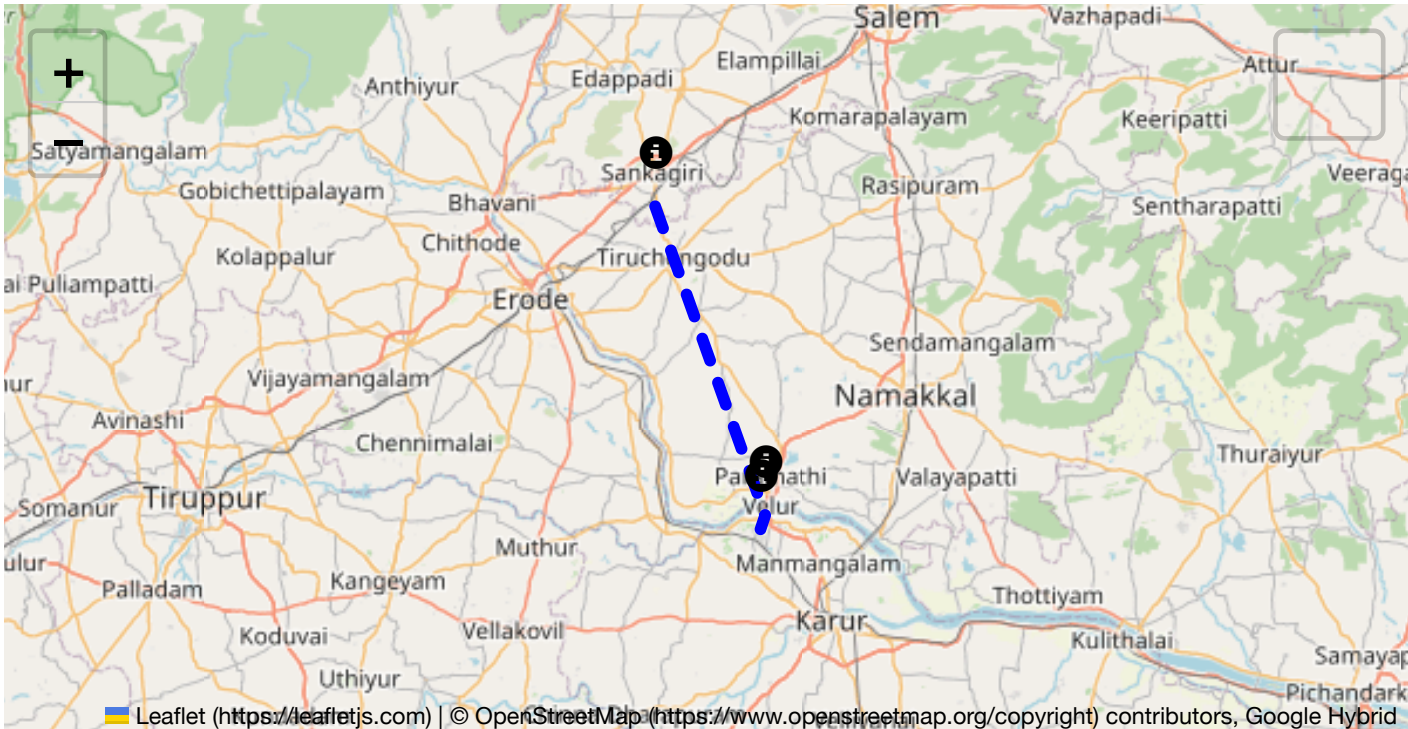
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

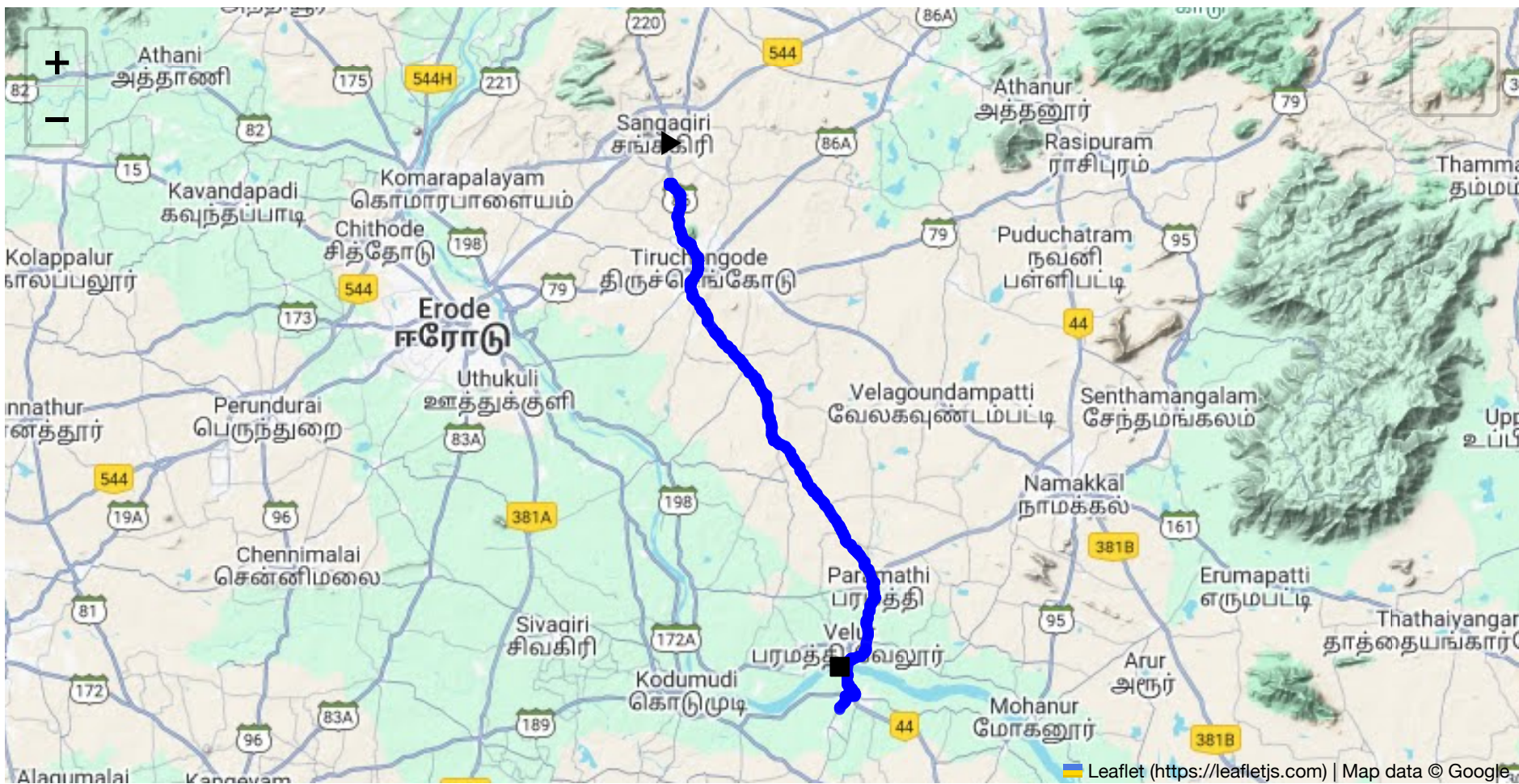
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 50.04 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours

Start: (11.4381, 77.8734)
End: (11.065724, 77.996663)



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to Ground Floor Punjaipugalur, Kagithapuram, via Chennis, Velayutham Palayam Rd, and Nanjaipugalur covers approximately 50.04 kilometers and typically takes about 1.08 hours for truck drivers transporting hazardous materials. This route travels through rural areas and small towns, with various road types interspersed between agricultural and forested regions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences hot, humid weather. The region is subject to monsoons, predominantly from October to December, which can result in heavy rain and potential flooding on roads. During peak summer (April to June), high temperatures might impact road surface conditions, potentially causing tar melting in poorly maintained areas.

3. Analysis of Traffic Patterns

Traffic along this route typically sees increased congestion during morning (6:30 AM - 9:00 AM) and evening (4:30 PM - 7:00 PM) hours, particularly near towns like Sangagiri and Kagithapuram, owing to local commuting. Agricultural equipment and slower-moving vehicles often traverse rural areas, which could cause delays.

4. Assessment of Road Quality and Infrastructure

Overall, the road quality varies. State highways are generally well-maintained, but secondary roads, such as those near Nanjaipugalur, may have potholes and uneven pavements. Bridges and crossings need

careful attention, as some may not be designed for heavy truck loads carrying hazardous materials.

5. Suggestions for Alternative Routes in Emergencies

In case of emergencies, alternative routes include:

- Taking the NH44 for a more direct path towards Karur and diverting towards the destination. It’s well-maintained but longer.
- Utilize local bypass roads around heavily congested towns.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires adherence to guidelines set by the Tamil Nadu Pollution Control Board, including valid permits and display of appropriate placards. Restrictions apply concerning timings; transportation during peak hours might need special permissions.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While exact records for the specific route are limited, reports from the region indicate occasional incidents due to overturned or skidded vehicles during monsoon seasons. Regulatory updates have improved oversight, but vigilance is still required.

8. Environmental Considerations and Sensitive Areas

The route crosses agricultural lands and several small water bodies, which calls for cautious driving to prevent contamination. Proximity to protected lands necessitates adherence to emission norms.

9. Analysis of Communication Coverage

Mobile network coverage is generally reliable, with potential dead zones near more rural, forested parts. Coverage can be patchy around Nanjaipugalur, hence, pre-trip checks on network maps for specific service providers are advisable.

10. Estimated Emergency Response Times for Different Route Segments

Emergency response times will vary depending on the segment:

- Sangagiri and Chennis: Approximately 30-45 minutes due to closer distance to larger towns.
- Velayutham Palayam Rd and Nanjaipugalur: Approximately 45-60 minutes due to rural nature and less immediate access.

12. Overall Summary of Risk Assessment

The route presents moderate risk primarily due to variable road conditions, local traffic patterns, and potential weather impacts. Regular updates on weather and traffic, adherence to local regulations, strategic planning of travel times, and consistent communication will mitigate most risks. Continuous monitoring of vehicle conditions, and applying best practices for transporting hazardous materials are crucial.

This analysis is aimed at ensuring safe and efficient travel, and helps equip drivers with relevant knowledge to manage potential hazards across the route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16755, 78.01568	30 KM/Hr
8	Turn	High	11.15573, 78.02176	15 KM/Hr
9	Turn	Medium	11.15565, 78.02171	30 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Blind Spot	Blind Spot	11.14480, 78.02291	10 KM/Hr
12	Blind Spot	Blind Spot	11.09003, 78.00251	10 KM/Hr
13	Blind Spot	Blind Spot	11.09005, 78.00260	10 KM/Hr
14	Turn	Medium	11.08001, 78.00539	30 KM/Hr
15	Blind Spot	Blind Spot	11.07467, 78.00842	10 KM/Hr
16	Blind Spot	Blind Spot	11.07476, 78.00192	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Velauthapalayam	11.0729521, 78.0023123	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16755, 78.01568



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15565, 78.02171



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.09003, 78.00251



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.09005, 78.00260



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.08001, 78.00539



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.07467, 78.00842



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.07476, 78.00192

